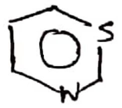
Ataur sir lecture:

Elemental analysis to identify unknown organic compounds by chemical methods.

Let,



Thiopyridine

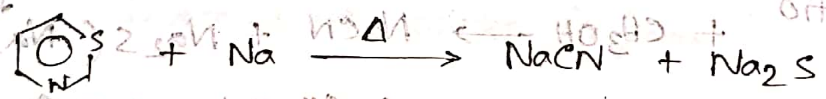
[যেমন, C, H, S, N element আছে। Halogen ও থাকতে পারে। অন্য compound এ।] S, N, X আছে কিনা তা test করবো। C, H test করবো না কারণ প্রায় সব organic compound এ থাকে, ডিফল্ট ধরে টেস্ট করবো]

S, -C, N-C, X-C molecule এ সব covalent bond এ যুক্ত

⇒ Na দ্বারা fusion করি কেন?

organic molecule is covalent compound so, it isn't water soluble, whereas the reagent that we are using are ionic compounds, and they are water soluble, So we need to make it an ionic compound. The elements are highly electronegative and we need highly electropositive metal to be happen. ওই fusion করে ionic compound করি।

Ex:



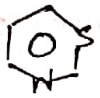
যেমন, অণু এমনভাবে ভাঙতে হবে যেন completely bond dissociation energy overcome করতে পারে, তাহলে, Na, S, H, C কেবলই আলাদা হয়ে যাবে। এরা সবাই highly reactive যেমন।

⇒ Na ই কেন use করি:

যদিও Na থেকে K more electropositive এবং এতে fusion ও অনেক ভালো হয়, কিন্তু safety issue বিবেচনা করে K ব্যবহার করা হয় না। Li এর size ছোট mostly covalent bond form করে তাই Li use করি না।

Fusion এ উৎপন্ন products:

(অক্সিজেন কম থাকলেও O_2 থাকলে)


$+ Na \xrightarrow{\Delta} NaCN + Na_2S + NO_2 + Na_2O + SO_2$

$H_2O + CO_2 + \overset{\times}{[CO_2]} + C + (\text{unreacted } Na) + NaS$

(O_2 উপস্থিত যুক্ত হয়) H_2 থেকে

[CO_2 এ convert হয় না)

(unreacted) black

[Na এর পরিমাণ কম হলে তখন S, N এর জন্য NaSCN ও form করবে]

⇒ unreacted Na এর কি হবে দেখি।

$Na + O_2 \longrightarrow Na_2O + \overset{\text{(steam)}}{H_2O} \longrightarrow NaOH$

steam $(H_2O) + Na \longrightarrow NaOH + H_{2(g)}\uparrow$

সমাধান : কেন : → [যদি Na থাকে বিস্ফোরণ ঘটবে]

Fused reactio medium (excess Na) $+ CH_3OH \longrightarrow NaCN + Na_2S$ ([কখনও কখনও] $NaSCN$) $+ NaOH + CH_3ONa + H_2\uparrow$

[Sodium methoxide]

(বিস্ফোরণ)

add H_2O before methanol → $NaOH + Na_2O$

(ইথানল বিক্রিয়া করে, অসুবিধা হয়.)

তাপমাত্রা কম হলে Na_2O বেশি হলে $NaOH$.

ছেঁকে filter → stock solution (যেটা দিয়ে যৌগের মধ্যস্থ বিভিন্ন element identify করা হয়]

① Sulfer test:

reaction medium এ ক্ষারীয়

$Na_2S + NaOH \xrightarrow{(CH_3COO)_2Pb}$ [$PbS\downarrow$ (black) , $Pb(OH)_2$ (white)] } intermediate brown colour.

$+$ other

↓ [যা detect করতে সমস্যা। তাই acid দিয়ে treat করতে হবে]

but Na_2S যে reagent এর সাথে reaction করবে অন্যগুলো তার সাথে reaction করবে না)

⇒ কেন CH_3COOH acid use করি HCl/HNO_3 ব্যবহারে কী অসুবিধা?

$$\underbrace{Na_2S + NaOH + HCl}_{\text{Neutral করে দেয়}} \xrightarrow{(CH_3COO)_2Pb} \underbrace{PbCl_2\downarrow(\text{white}) + PbS\downarrow(\text{black})}_{\text{Again intermediate colour.}}$$

⇒ HNO_3 use করলে $Pb(NO_3)_2$ তৈরি হয় যেটাও সাদা তাই use করি না।

[$H_2SO_4 \rightarrow PbSO_4$ white]

⇒ Acetic acid এর সুবিধা?

$$Na_2S + NaOH + CH_3COOH \xrightarrow{Pb(CH_3COO)_2} (CH_3COO)_2Pb + \underset{\text{(intact)}}{CH_3COOH} + \underset{\text{Sharp black}}{PbS\downarrow}$$

[Here no intermediate colour is formed]

[এখানে $Pb(CH_3COO)_2$ এর corresponding anion আছে CH_3COO^-]

(ii) Nitrogent test:

$$NaCN + FeSO_4 \longrightarrow Na_2SO_4 + Fe(CN)_2$$

(no green p.p.t)

যদি ppt bluish না হয় তখন ~~NaOH~~ আবার NaOH add করতে হবে। তার মানে stock solⁿ এ NaOH ছিল না।

① Sulfer test:

reaction medium এ থাকিবে

$$Na_2S + NaOH \xrightarrow{(CH_3COO)_2Pb} \left[\begin{array}{l} PbS\downarrow \text{ (black)} \\ Pb(OH)_2 \text{ (white)} \end{array}\right\} \text{intermediate brown colour.}$$

$+$ other

↓ [এর detect করতে সমস্যা । তাই acid দিয়ে treat করতে হবে]

[but Na_2S যে reagent এর সাথে reaction করতে যাচ্ছিল তার সাথে reaction করবে না]

⇒ কেন CH_3COOH acid use করি HCl/HNO_3 ব্যবহারে কী অসুবিধা?

$$Na_2S + NaOH + HCl \xrightarrow{(CH_3COO)_2Pb} \underbrace{PbCl_2\downarrow\text{(white)} + PbS\downarrow\text{(black)}}_{\text{Again intermediate colour.}}$$

Neutral করে দেয়

⇒ HNO_3 use করলে $Pb(NO_3)_2$ তৈরি হয় যেটা কালো আনে use করি না।

[$H_2SO_4 \rightarrow PbSO_4$ white]

⇒ Acetic acid এর সুবিধা?

$$Na_2S + NaOH + CH_3COOH \xrightarrow{Pb(CH_3COO)_2} (CH_3COO)_2Pb + \underset{\text{(intect)}}{CH_3COOH} + \underset{\text{sharp black}}{PbS\downarrow}$$

[Here no intermediate colour is formed]

[যেহেতু $Pb(CH_3COO)_2$ এর corresponding anion আছে CH_3COO^-]

(ii) Nitrogent test:

$$NaCN + FeSO_4 \longrightarrow Na_2SO_4 + \underset{}{Fe(CN)_2}$$

[no green p.p.t]

যদি ppt bluish না হয় তখন ~~NaOH~~ আবার NaOH add করতে হবে । তার মানে stock solⁿ এ NaOH ছিল না।

কেন NaOH দিবো?

Bluish ppt. না হলে NaOH দিব। ppt. bluish হলে NaOH দেওয়ার দরকার নেই।

$NaCN + FeSO_4 + NaOH \text{ (1drop) extra} \longrightarrow \overset{\text{Greenish p.pt.}}{Fe(OH)_2} + NaCN + Na_2SO_4$

(যেন এটা form হয়)

$Fe(OH)_2 + NaCN \longrightarrow Fe(CN)_2 + 4NaCN$

$\downarrow$

$Na_4[Fe(CN)_6]$

$Na_4[Fe(CN)_6] + FeCl_3 \longrightarrow Fe_4[Fe(CN)_6]_3 \downarrow + NaCl$

(prussian Blue)

[Ferric ferro cyanide]

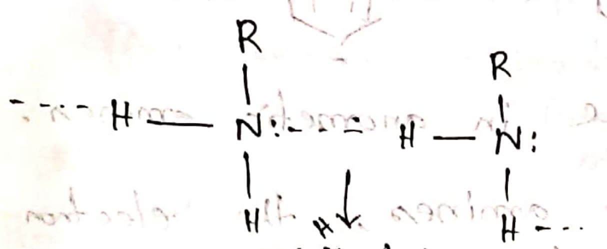
KF use করি কেন?

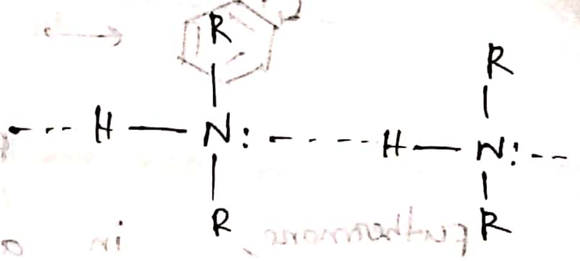
যদি কোন কারণে Fe^{3+} এর পরিমাণ বেশি হয় তখন Fe^{3+} এর colour yellow. আর colour interupt করে। green colour হয়।

$KF + Fe^{3+}Cl_3 \longrightarrow KFeCl_4$

$\text{(yellow)} \qquad \text{(colourless)}$

Solubility of amines:





This bond is weaker than the O---H bond because of electronegativity.

This bond is weaker than 1° amine because -R group can stabilizes the intensity of N-atom, becomes less electronegative.

(B.P → 1° amine > 2° amine > 3° amine
1° amine এ 3টি H-bond form করে যা 2° তে 2টি 3° এ 1টি H-bond form করে। বেশি H-bond মানে বেশি ~~bond~~ ~~strong~~ bond ভাঙতে হবে]

H-bond তো বেশি হয় solubility তত বেশি হবে। সুতরাং:
1°>2°>3° solubility R এর সাথে ইহা depend করে। যেহেতু R group [non-polar]

Basicity of amines:

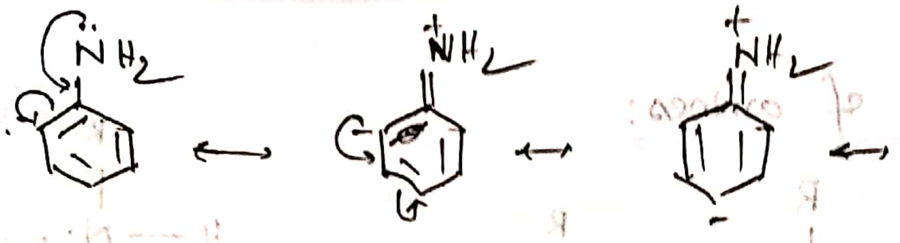
$$\overset{2^\circ}{(CH_3)_2NH} > \overset{1^\circ}{CH_3NH_2} > \overset{3^\circ}{(CH_3)_3N} > NH_3$$

[Depend on the stability of conjugate acid]
~~[illegible]~~ এ উপর [illegible] [illegible]

The basicity order:

~~Aniline~~ Aromatic amine is less basic than aliphatic amine:

In case of aromatic amines the lone pair electron of Nitrogen are not always available (to acept proton) due to effect of resonance.



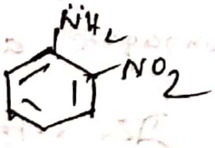
Resonance in aromatic amines.

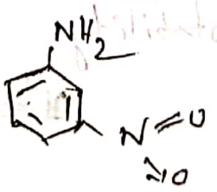
Futhermore, in aliphatic amines, the electron density of N atoms is increased by electron donating inductive effect. Therefore the pair of nitrogen atom is easily available for donating on protonation.

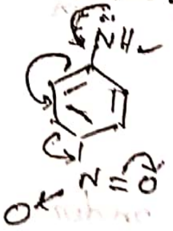
The basicity order is,

Aniline > meta-nitroaniline > para-nitroaniline > ortho-nitro aniline.

Electron withdrawing group can decrease the electron density of aniline. In aniline there has no electron withdrawing group. Therefore it has high electron density. So more basic than nitroaniline.







ortho and para position এ resonance effect এ basicity কমায় but meta position এ resonance effect নেই।

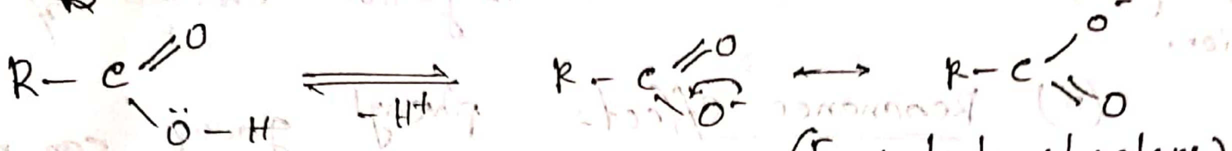
(ortho position এ থাকার কারণে plane এ থাকে না -NO$_2$ group তাই resonance তৈরি হয় না) → but closer তাই inductive effect powerful।

carboxylic acid is more acidic than alcohol:

$$R-OH \longrightarrow RO^{-} + H^{+}$$

[Alcohol unionized] [Alkoxide ion]

[unstable there has no resonance structure]

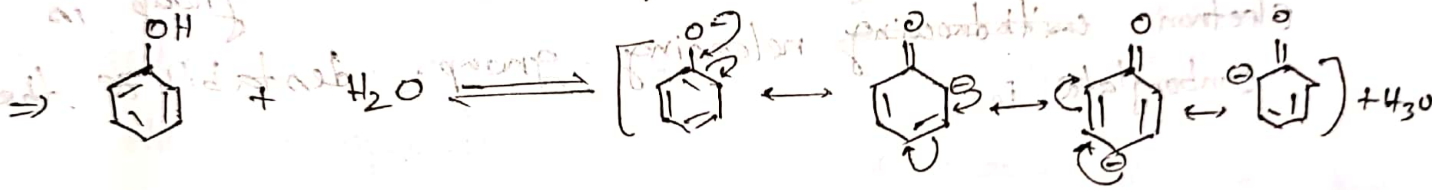


(Equivalent structure)

(Resonance is more important)

⇒ carboxylic acid is more stronger acid than phenol?

⇒ resonance form of carboxylic acid: ⟶



⇒ in carboxylate ion the negative charge is effectively delocalized between more electronegative oxygen atoms.

⇒ For phenoxide, negative charge is less effectively delocalized over one oxygen atom and less electronegative carbon atoms.

⇒ Aromatic carboxylic acids is more acidic than Aliphatic carboxylic acid?

(Resonance চিত্রে আছে)

(In aliphatic –COOH group attached with sp^3 hybridized carbon atom) In aromatic sp^2 hybridized carbon atom.

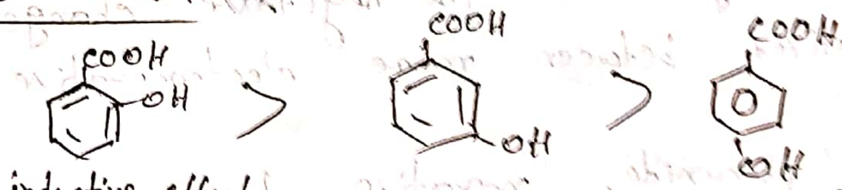
In aromatic carboxylic acid there has two factor:

(i) Inductive effect:- sp Electron withdrawing phenyl group can stabilize the negative charge of the anion.

(ii) Resonance effect:- phenyl group can release electron and destabilized the carboxylate ion and decrease acidity.

⇒ In aliphatic carboxylic acid, the alkali group is electron ~~withdrawing~~ releasing group. destabilize the carboxylate ion.

⇒ Acidity order:

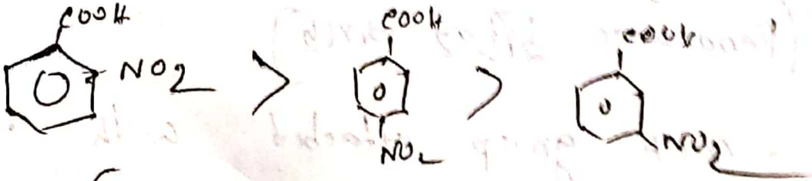


inductive effect so effective, resonance is less effective, plane ও [illegible] repulsion ও থাকে। (intramolecular H-bond form করে ও [illegible] [illegible] [illegible] [illegible])

(inductive effect less Resonance effect less effect than o-)

(inductive effect is less effective resonance effect is effective)

inductive effect e⁻ টানে
resonance " " ছাড়ে



(pKa কম হলে বেশি acidic)

Tutorial examination:

2020

① Why is metallic Na used in fusion of organic compounds? Why some methanol is used after fusion? Write the name of compounds formed upon decompositio with Na → 3/.

② How nitrogen can be identified in unknown compounds? Write the necessary reactions → 2.5

(3) An organic compound is soluble in NaOH but insoluble in $NaHCO_3$. What is the solubility class of this compounds? Mention the name of probable compounds of this class — (2.5).

(4) What is the tollens reagent? How aldehyde and ketones can be ~~identified~~ differentiated using it? → (2).

(5) An organic compounds shows IR bands at 3380 and 1690 cm^{-1}. It produce brisk effervescence upon addition of dil HCl and 20% $NaNO_2$ rather reacting with 2,4-DNP to form red dye. Assume the class of compound and give necessary explanation. — (3).

(6) What is meant by derivatives: Write the name of two derivatives for each of the following compounds (i) acid
(ii) carbonyl compound.
—(3).

(7) predict the structure of the compound have the molecular formula $C_4H_8O_2$ with the help of following spectroscopic data:

IR: $\nu = 2976, 2855, 1740, 1202 \, cm^{-1}$

NMR : $\delta = 3.7$ (3H, s), 2.4 (2H, q), 1.2 (3H, t) ppm.

2020

① Why is the sodium extract alkaline in nature? —(2)

(2) What is prussian blue test? write down the chemical equation for this test —(3).

(3) Mention the solubility class of the following compounds: —(3).

(a) Benzophenone (b) Benzoic acid (c) D-Glucose
(d) Amino phenol (e) Napthalene (f) 4-methylaniline

(4) How can you differentiate between the following pairs by chemically?? Give necessary equations —(4).

(a) Aniline and methylamine

(b) Hexanol and 3-hexanone

⑤ What is the tollent reagent! How can you prepare this in the laboratory?

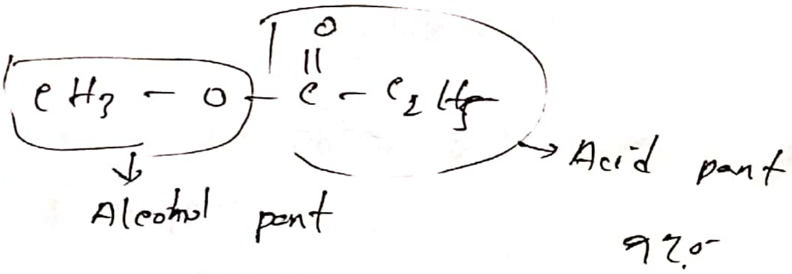
—(2)

(6) Write the expected IR and H-NMR spectra value of the following compounds..

(a) Acetamide —(6)

(b) Ethyl acetate.

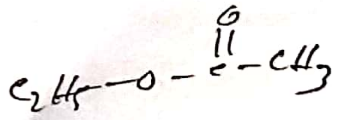
ester ഉ: [illegible]


$CH_3 - O - \overset{O}{\overset{\|}{C}} - C_2H_5$
Acid part
Alcohol part

920

[illegible]

methyl ethanoate

ethylethanoate → 

$C_2H_5 - O - \overset{O}{\overset{\|}{C}} - CH_3$

[3° amine এর IR এ কোনো $\ddot{N}-H$ peak পাব না। N-H bond নেই]

⇒ What is sodium fusion test:/Lassaignes test.

⇒ The sodium fusion test is used ~~to~~ in elemental analysis for the quantitative determination of elemental halogen, nitrogen and sulfur in a sample/in a organic compound.

$$C, N, S, X \xrightarrow{Na,\ heat} Na_2S, NaCN, NaCNS, NaX$$

$X = [Cl, Br \text{ and } I]$

⇒ Detection of sulfur:

(i) Lead acetate test:

$$Na_2S + (CH_3COO)_2Pb \xrightarrow{dil.\ CH_3COOH} PbS\downarrow + 2CH_3COONa$$

(black or brown ppt)

---

dil. CH_3COOH is added to this solution instead of H_2SO_4. If H_2SO_4 is used, $Pb(CH_3COO)_2$ will itself react with H_2SO_4 to form white ppt of lead sulphate.

$$\underset{\text{lead acetate}}{Pb(OOCCH_3)_2} + H_2SO_4 \longrightarrow \underset{\text{(white ppt.)}}{PbSO_4\downarrow} + 2CH_3COOH$$

Hence the white ppt of $PbSO_4$ will interfere with the following test of sulphur.

$$Pb(CH_3COO)_2 + Na_2S \longrightarrow \underset{\text{(Black ppt)}}{PbS\downarrow} + 2CH_3COONa$$

However if acetic acid is used, it does not react with lead acetate, so does not interfere in test.

CS CamScanner

Firstly the fuison filtnate is acidified with dil. CH_3COOH until the solution is acidic. Then if we add lead acetate, the solution containing the sulfer react to give the black ppt presence of PbS which indicated the presence of sulphur.

(ii) Sodium nitroprosside test:

If we add only a drop of 0.10% sodium nitroprosside to the fusion filtrate contains sulfur, the reaction will produce a deep red or purple colour solution.

$$Na_2S_{(aq)} + \underset{(SNP)}{Na_2[Fe(CN)_5NO]_{aq}} \longrightarrow \underset{\text{Sodium Sulfonitroprosside}}{Na_4[Fe(CN)_5NOS]}$$

purple colour/deep red

Detection of nitrogen:

① prussian blue test:

$$NaOH + FeSO_4 \longrightarrow Fe(OH)_2 + Na_2SO_4$$

$$Fe(OH)_2 + 6NaCN \longrightarrow Na_4[Fe(CN)_6] + 2NaOH$$

Sodium ferrocyanide (blue)

$$3Na_4[Fe(CN)_6] + 4FeCl_3 \longrightarrow Fe_4[Fe(CN)_6]_3 + 2NaCl$$

Ferric ferro cyanide (prussian blue)

/Distin… …off's test: Mix … Heat the mixture to …
…ted in water. … the other hand, long standing …
…inal and HCl or concen…

CS CamScanner

The ~~prussian~~ prussian blue colour or blue precipitate indicates the presence of nitrogen.

How would you give a combine test for N and S?

⇒ If any organic compound contains N and S during the sodium fusion it may produce sodium thiocyanate (NaCNS). A fusion filtrate containing NaCNS is first acidify with dil. HCl, then a drop of freshly prepared 2.5% $FeCl_3$ will give a blood red colorati due to the formation of the compound ferric thiosyanochloride which indicates the simultaneous presence of N and S.

$$NaCNS + FeCl_3 \rightleftharpoons Fe(CNS)Cl_2 + NaCl$$

Ferric thiocyanochloride
(blood color)

Detection of Halogens:

If N and S is present add HNO_3 to acidify the solution filtrate by heating the reduce halide:

$$NaCN + HNO_3 \xrightarrow{\Delta} NaNO_3 + HCN\uparrow$$

$$Na_2S + HNO_3 \xrightarrow[\Delta]{\cancel{NaNO_3}} NaNO_3 + H_2S\uparrow$$

$$NaX + HNO_3 \xrightarrow{\Delta} NaNO_3 + HX$$

⇒ Why cyanide or sulphide ions must be removed before adding silver ions? to test for the halides?

⇒ The lassaigne sample, when selected to test for halides ions, if cyanide or sulphide ions are present in this solution:

$$AgNO_3 + CN^- \longrightarrow \underset{\text{(white)}}{\underset{\text{ppt}}{AgCN\downarrow}} + NO_3^-$$

$$AgNO_3 + S^{2-} \longrightarrow \underset{\text{(black)}}{\underset{\text{ppt}}{Ag_2S}} + NO_3^-$$

We see that if CN^- and S^{2-} ions are not removed, they will react with Ag^+ to result in positive halide test through the sample does not contain any halogen.

→ How to remove CN^- and S^{2-} ions?

We can remove CN^- ions and S^{2-} ions from the sample by adding conc. HNO_3 acid to it.

$$NaCN + HNO_3 \xrightarrow{\Delta} NaNO_3 + HCN\uparrow$$
$$Na_2S + HNO_3 \xrightarrow{\Delta} NaNO_3 + H_2S\uparrow$$

$NaNO_3$ being a soluble salt does not interfare further [HCN and H_2S can escape from the solⁿ]

HNO_3 ছাড়া অন্য কোনো acid use করা যায় না তাহলে ppt পড়বে। H_2SO_4 use করলে Na_2SO_4 এর ppt পড়বে।

⇒ Why the sample after being acidified with conc. HNO_3 need to boiled?

⇒ The removal of HCN/H_2S made from the reaction of CN^- and S^{2-} ions with HNO_3 is effected by boiling the fusion solution to reduce it to half of its initial volume.

The sample after being acidify with conc. HNO_3 to remove HCN and H_2S to boil.

Reaction with $AgNO_3$:

$$NaCl + AgNO_3 \longrightarrow \underset{\text{(white)}}{AgCl} + NaNO_3$$

$$NaBr + AgNO_3 \longrightarrow \underset{\text{(pale yellow)}}{AgBr} + NaNO_3$$

$$NaI + AgNO_3 \longrightarrow \underset{\text{(yellow)}}{AgI} + NaNO_3$$

⇒ **Differentiation between Cl, Br, I :-**

(i) **Reaction with HNO_3 (nitric acid):**

$$NaCl + HNO_3 \xrightarrow{CCl_4} NaNO_3 + HCl$$

$$HCl + HNO_3 \longrightarrow Cl_2 \text{ (in } CCl_4 \text{ colourless)}$$

CS CamScanner

$$NaBr + HNO_3 \xrightarrow{CCl_4} NaNO_3 + HBr$$

$$HBr + \underset{\text{(oxidizing agent)}}{HNO_3} \longrightarrow Br_2 \text{ (in } CCl_4 \text{ deep brown)}$$

$$NaI + HNO_3 \xrightarrow{CCl_4} NaNO_3 + HI$$

$$HI + \underset{\text{(oxidizing agent)}}{HNO_3} \longrightarrow I_2 \text{ (in } CCl_4 \text{ violet)}$$

CamScanner

Class S_A :

Ether and H_2O soluble acid, monofunctional carbonylic acid. (can form H-bond with ~~ether and~~ H_2O).

⇒ ~~Ether can form hydrogen bonds to water, since the oxygen atom is attracted to the partially-positive hydrogen in water molecules, making them more soluble in water than alkanes.~~

$CH_3-O-CH_2-CH_3 \rightarrow$ soluble

$(CH_3-CH_2-CH_2-CH_2)O \rightarrow$ insoluble

ছোট ether H-bond form করে তাই water soluble but বড় ক্ষেত্রে ether H-bond form করতে পারে না তাই insoluble

⇒ Acetic acid is soluble both water and ether.

Acetic acid has a polar group with the $-COOH$ at one end. polar substances readily from H-bond in a polar molecule (liquid) such as water. Diethyl ether is also somewhat polar, so that's part of it.

The other end of acetic acid is $-CH_3$ which is non-polar. This part dends to mix nicely with other non-polar staff such as the ethyl parts of the ether.

① Reaction of primary (1°) aliphatic amines with nitrous acid :—

produce of HNO_2 acid (Highly unstable).

$$\underset{\text{(Sodium nitrite)}}{NaNO_2} + HCl \xrightarrow{0-5^\circ c} HNO_2 + NaCl$$

$$\underset{\text{primary amine}}{CH_3-CH_2-CH_2-NH_2} \xrightarrow[0-5^\circ c]{NaNO_2 + HCl} CH_3-C_2H_4-\overset{+}{N}\equiv N\,Cl^- + H_2O + Na^+ + c$$

কার্বোক্যাটায়ন ই বোঝায়

Aliphatic diazonium salt
highly unstable

$$\underset{\text{(1° carbonium ion)}}{CH_3-CH_2-\overset{+}{C}H_2} + \underset{\text{brisk effervescence}}{N_2} + Cl^-$$

highly unstable

বুদবুদ

$$CH_3-\underset{H}{CH}-\overset{+}{C}H_2 \xrightarrow{\text{Elimination}} CH_3-CH=CH_2 + H^+$$

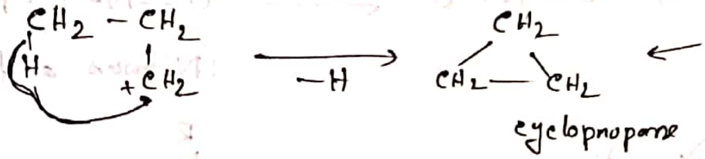
$$CH_3-CH_2-\overset{+}{C}H_2 + Cl^- \longrightarrow \underset{\text{n-propyl chloride}}{CH_3-CH_2-CH_2Cl}$$

$$CH_3-\underset{H}{CH}-\overset{+}{C}H_2 \xrightarrow{\text{Rearrangement}} \underset{\text{2° carbonium ion}}{CH_3-\underset{+}{CH}-CH_3} \xrightarrow{Cl^-} \underset{\text{2-chloropropane}}{CH_3-\overset{Cl}{CH}-CH_3}.$$

$$CH_3-CH_2-\overset{+}{C}H_2 + H_2\ddot{O} \longrightarrow CH_3-C_2H_4-\underset{H}{\overset{\oplus}{O}}-H \xrightarrow{-H} \underset{\text{n-propanol}}{C_3H_7-OH}$$

~~n-heptanol~~.

$$\underset{\text{n-propanol}}{C_3H_7OH} + \underset{\text{1° carbonium}}{C_2H_5-\overset{+}{C}H_2} \xrightarrow[-H]{} \underset{\text{Ether}}{C_3H_7-O-C_3H_7}$$



← এটাও হয় stable, কারণ খুব low temp. reaction টা হয়।

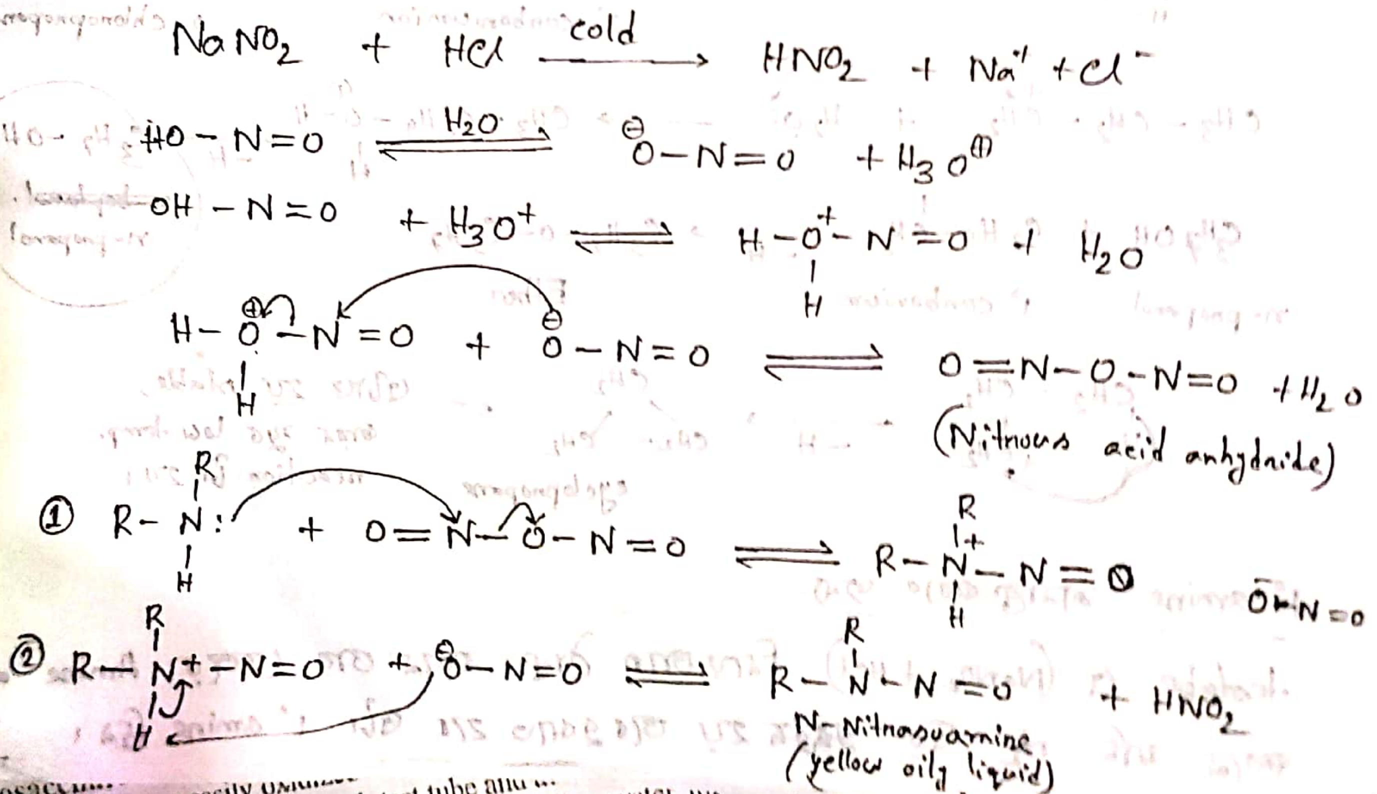
1° amine শনাক্ত করার উপায়:

testube এ ($NaNO_2$ + HCl) নিয়ে তাকে ঠান্ডা করে তার ভেতর Amine add করলে যদি N_2 গ্যাস বুদবুদ হয় তাহলে বুঝতে হবে এটা 1° amine ছিল।

2° amine শনাক্ত করার জন্য: কিছু উপাদানে যদি আমাদের 2° amine থাকে তার সাথে HNO_2 যোগ করি yellow oily liquid তৈরি হবে। টেস্টটিউবের মধ্যে উপরের দিকে দেখতে পাব। আরও conform হওয়ার জন্য তারপরেই যদি conc. H_2SO_4 দিয়ে heat করি (and কিছু phenol add করি) তখন যে ~~N~~-Nitrosonium ion তৈরি হবে, সেটা পরে phenol এর সাথে reaction করে HO-⟨⟩=N-OH হয় তখন তখন আবার sol^n এ H_2SO_4 add করলে colour green হবে, এরপর H_2O add করে dil. করলে red colour হবে। এটাকে আবার base এর সাথে reaction করে salt form করলে সেটার blue colour হবে।

3° amine শনাক্ত করার জন্য: clear sol^n হবে HNO_2 add করলে। oil তৈরি করে না, N_2 gas তৈরি করে না। water soluble clear sol^n হয়।

(Reaction of 2° (aliphatic amines with nitrous acid)



$$NaNO_2 + HCl \xrightarrow{cold} HNO_2 + Na^+ + Cl^-$$

$$HO-N=O \overset{H_2O}{\rightleftharpoons} \overset{\ominus}{O}-N=O + H_3O^{\oplus}$$

$$OH-N=O + H_3O^+ \rightleftharpoons H-\overset{+}{O}(H)-N=O + H_2O$$

$$H-\overset{\oplus}{O}(H)-N=O + \overset{\ominus}{O}-N=O \rightleftharpoons O=N-O-N=O + H_2O$$

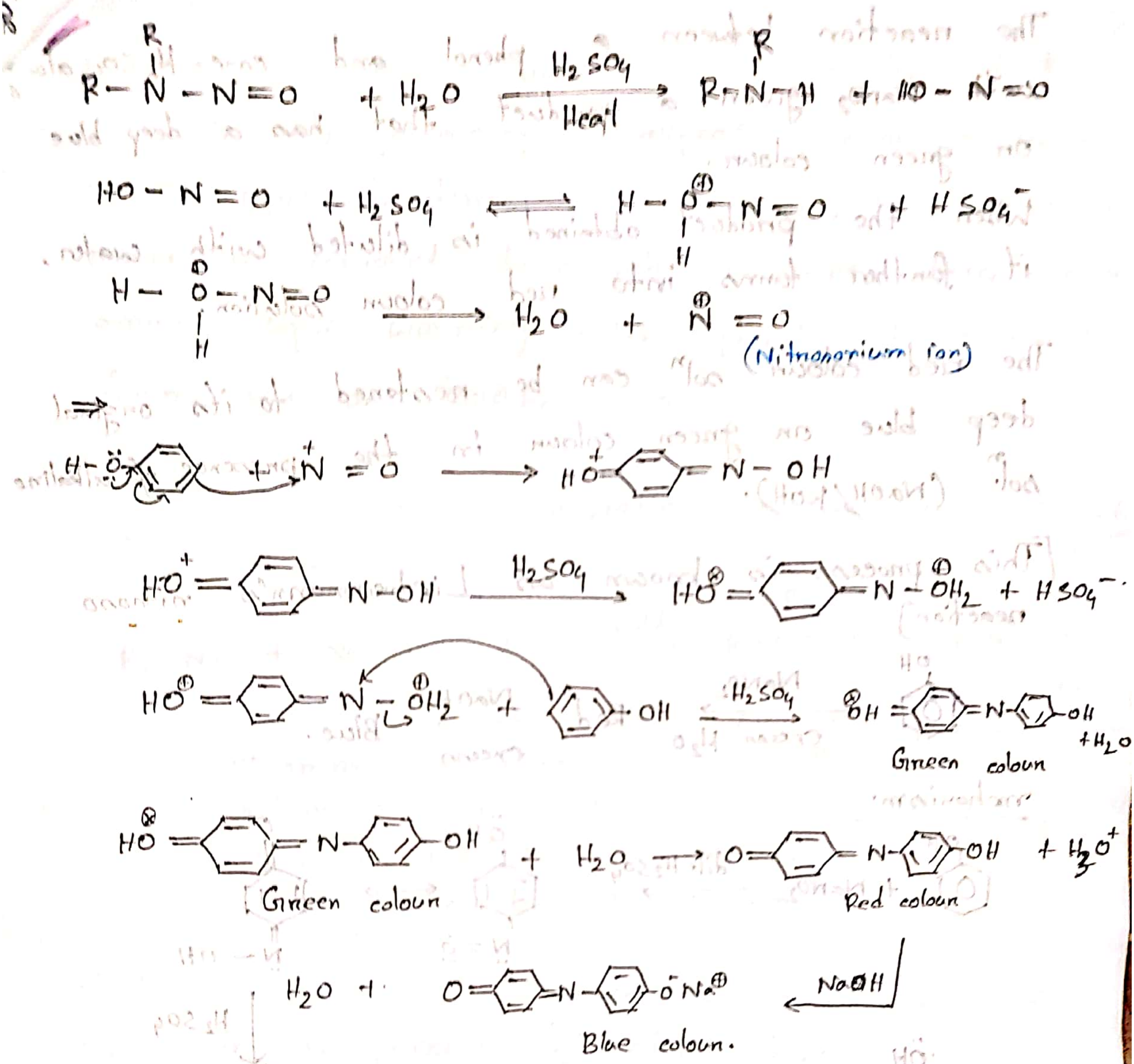
(Nitrous acid anhydride)

① $R-\ddot{N}(R)(H) + O=N-O-N=O \rightleftharpoons R-\overset{+}{N}(R)(H)-N=O \quad \overset{-}{O}-N=O$

② $R-\overset{+}{N}(R)(H)-N=O + \overset{\ominus}{O}-N=O \rightleftharpoons R-N(R)-N=O + HNO_2$

N-Nitrosoamine
(yellow oily liquid)

onosacc
ides are easily oxid
inch test tube and
boiling water



$$R-\overset{R}{\overset{|}{N}}-N=O + H_2O \xrightarrow[\text{Heat}]{H_2SO_4} R-\overset{R}{\overset{|}{N}}-H + HO-N=O$$

$$HO-N=O + H_2SO_4 \rightleftharpoons H-\underset{H}{\underset{|}{\overset{\oplus}{O}}}-N=O + HSO_4^-$$

$$H-\underset{H}{\underset{|}{\overset{\oplus}{O}}}-N=O \longrightarrow H_2O + \overset{\oplus}{N}=O$$

(Nitrosonium ion)
H₂SO₄
Green colour
+H₂O
Green colour
Red colour
NaOH
Blue colour.

"Liebermann's nitroso reaction" ✓

⇒ Liebermann's nitrose reaction can be used as a test to distinguish secondary (2°) aliphatic or aromatic amines containg a nitroso group.

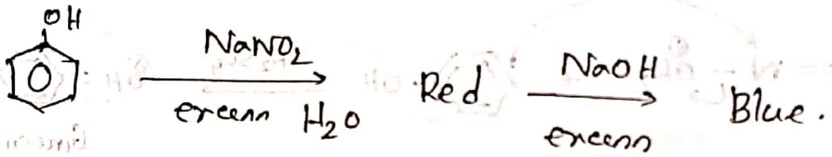
(Liebermann's reagent is composed of 10mL of conc. H_2SO_4 and 1g of KNO_2.)

The reaction between a phenol and conc. H_2SO_4, along with $NaNO_2$ gives a product that has a deep blue or green colour.

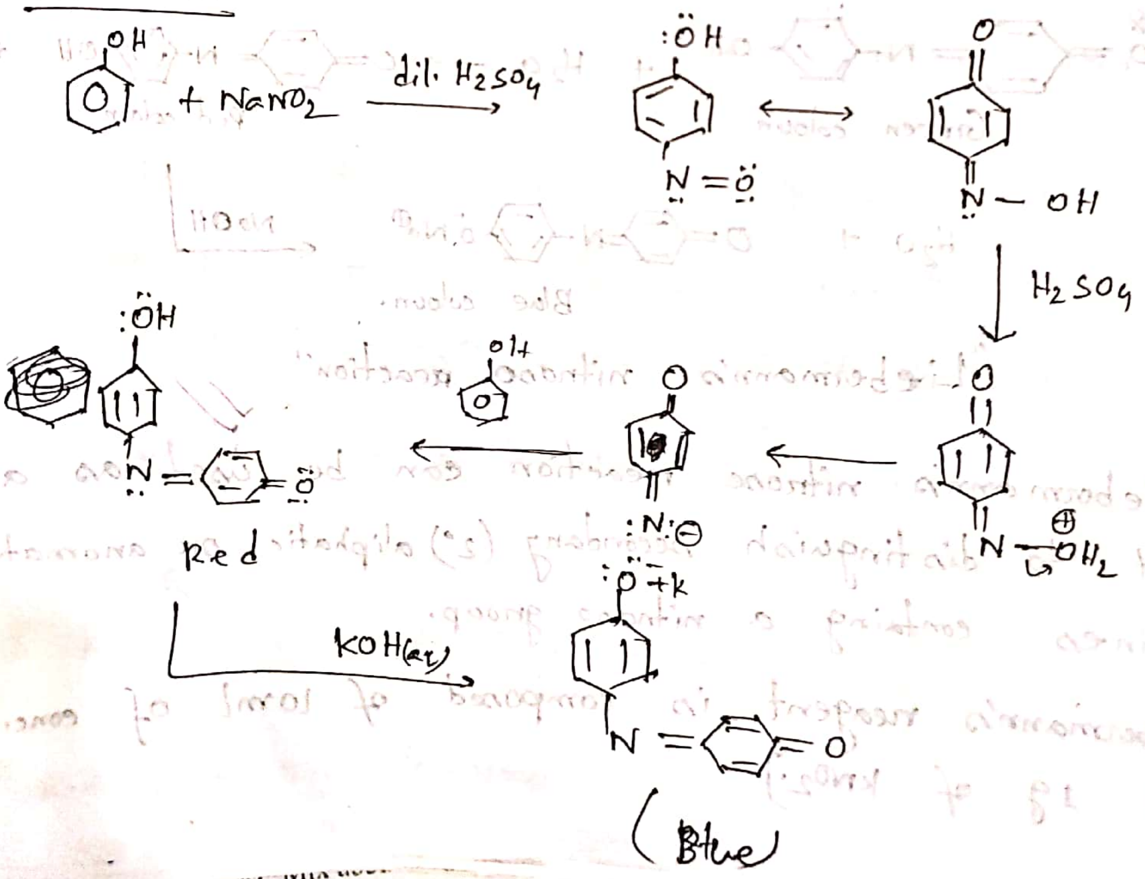
When the product obtained is diluted with water, it further turns into red colour solution.

The red colour soln can be restored to its original deep blue or green colour in the presence of alkaline soln ($NaOH/KOH$).

[This process is known as Liebermann's nitroso reaction]


OH
$NaNO_2$
excess H_2O
Red
NaOH
excess
Blue.

mehanism:


OH
+ $NaNO_2$
dil. H_2SO_4
:ÖH
N=Ö:
O
N—OH
H_2SO_4
O
N—OH_2 ⊕
O
:N:⊖
OH
:ÖH
N=
Ö:
Red
KOH(aq)
:Ö⁻+K
N=
=O
(Blue)

Seliwanoff's test. Mix ... tested in water. Heat the mixture ... other hand long standing or ...

Libermann's nitroso reaction : it is a test that is give only secondary (2°) aliphatic amine or aromatic amine. upon treatement with (HNO_3), the 2° aliphatic amine is converted to a nitroso amine. This nitroso amine, upon warming with phenol and conc H_2SO_4 form red or ~~blue~~ brown colour.

Reaction of tertiary 3° aliphatic amines: with nitrous acid.

2° amine হলে এখানে H থাকে

$$R_3-\ddot{N} + NaNO_2 + HCl \xrightarrow{cold} \underset{\substack{\text{Aminium}\\ \text{chloride}}}{R-\overset{R}{\underset{R}{\overset{|+}{N}}}-HCl^-} + \underset{\substack{\text{N-Nitrosoammonium}\\ \text{chloride}\\ \text{colourless.}}}{R-\underset{R}{\overset{R}{N^{\oplus}}}-N=O\,Cl^-} + H_2O$$

3° amine এর সাথে nitrous acid clear sol^n হয়ে যায়: oil হয় না, N_2 gas হয় না; water soluble clear sol^n হয় (salt).

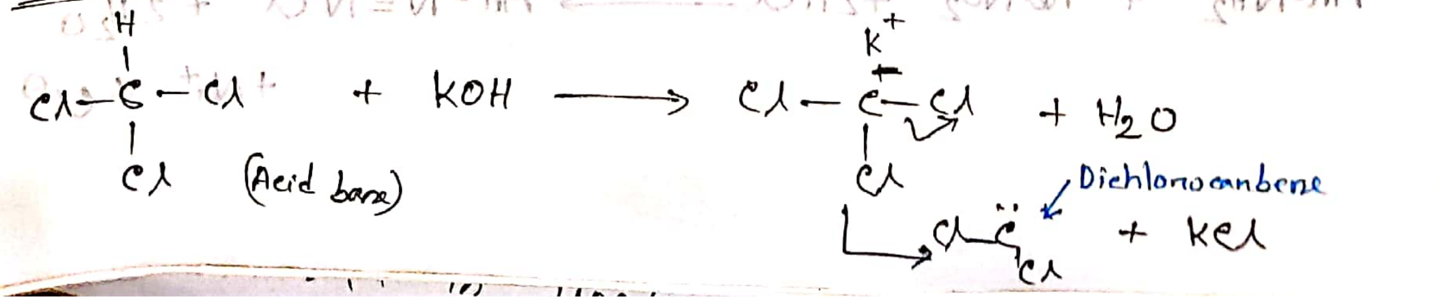
⇒ Reaction of aliphatic/aromatic amins with chloroform in the presence of strong base.

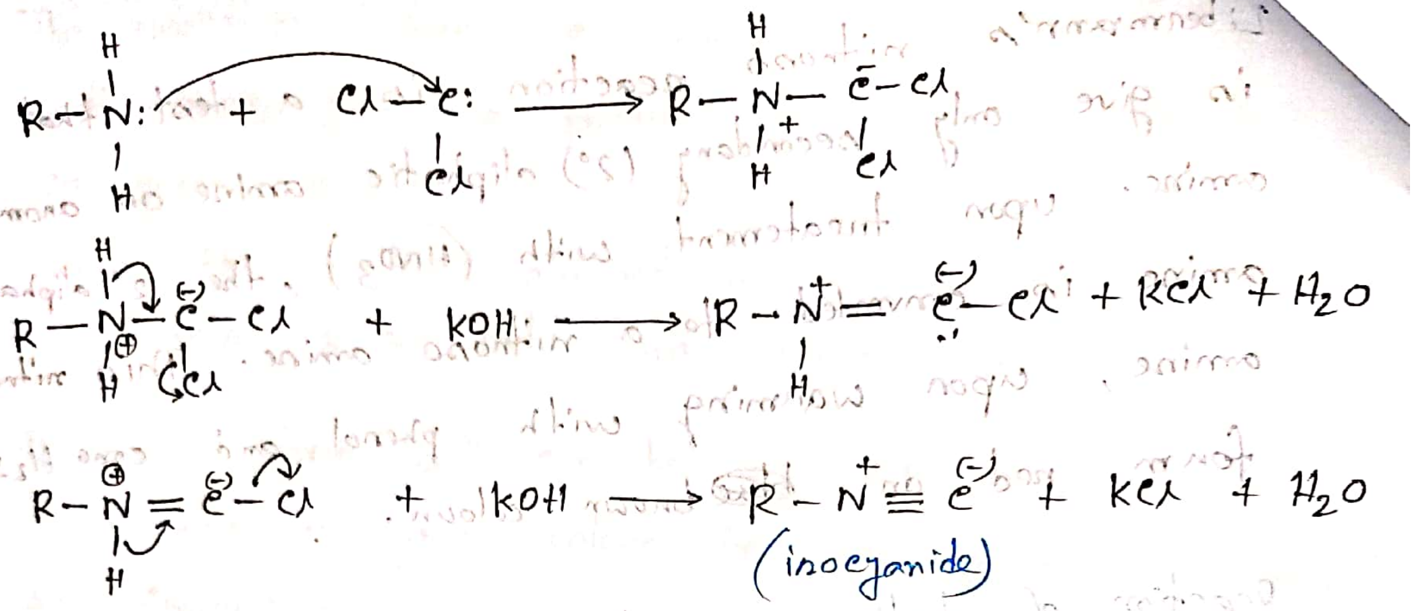
① carbylamine (isocyanide test):

[এটা উৎকট গন্ধ হয় সমস্ত গন্ধ যেন reaction stop করে দিয়েছে]

$$R/Ar-NH_2 + CHCl_3 + 3KOH \xrightarrow{heat} \underset{\text{Alkylisocyanide}}{R/Ar-\overset{+}{N}\equiv\overset{-}{C}} + 3KCl + H_2O$$

mechanism:



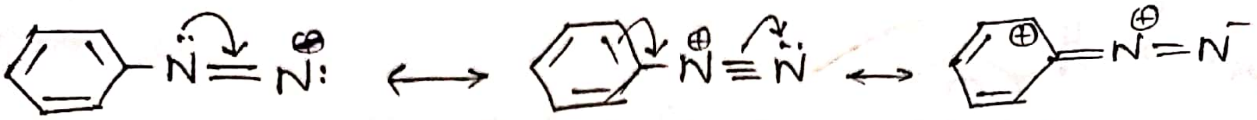


~~$R-N=C-Cl$~~ carbilamine reaction involves the formation of isocyanides or carbylamines by the addition of the aromatic and aliphatic primary amines on heating with chloroform and ethanoic potassium hydroxide. The formed carbylamine or isocyanides will have foul smell. This reaction is called carbylamine test.

Aliphatic or aromatic primary amine সমস্ত কেবল শুধু এই reaction দেখায় তার হয়। অন্যসব, 2° বা 3° amine ক্ষেত্রে H না থাকায় এই reaction হয় না।

Reaction of primary aryl amines with nitrous acid:

$$Ar-NH_2 + NaNO_2 + 2HCl \xrightarrow{0-5^\circ C} Ar-\overset{\oplus}{N}\equiv N\,Cl^- + 2H_2O + Na^+ + Cl^\ominus.$$

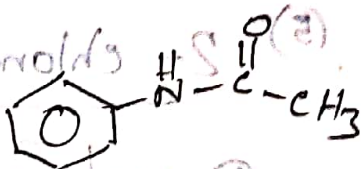


[Resonance stabilized aromatic diazonium ion.]

⇒ Nitrous এসিড ৰ সহায়ত আমি aromatic primary amine and aliphatic primary amine ৰ diff. কৰিব পাৰি।

⇒ The diazonium salt of aliphatic primary amine is not resonance stabilized. Therefore this diazonium salt gives a brisk effervescence of nitrogen gas. On the other hand, the aromatic diazonium salt is resonance stabilized. That can not be broken. কাৰণ, C=N double bond are produced by resonance So that cannot give us N_2 gas. এতেকে এই তাপমাত্ৰাত (কম) aromatic diazonium salt stable হয় উচ্চ তাপমাত্ৰাত ভাঙি যায়।

① **Acetanilide:**



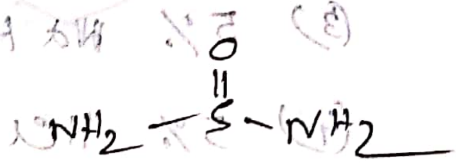
114°c

① water insoluble

② 5% HCl insoluble.

③ ~~N/S~~ present So miscellaneous.

② **Thiourea:**


$NH_2-\overset{O}{\overset{\|}{S}}-NH_2$

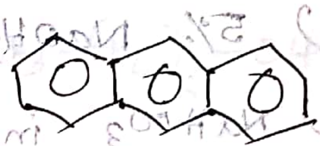
182°c

① water soluble

② ether insoluble (class)2

③ N/S present → Amino sulphonic acids

③ **Anthracene:**



215°c

① water insoluble

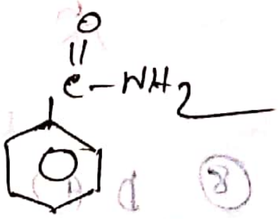
② 5% NaOH insoluble

③ 5% HCl insoluble (N,S not present)

④ conc. H_2SO_4 soluble.

⑤ 85% H_3PO_4 insoluble class $-N_2$

④ **Benzamide:**



127-130°c

① water insoluble.

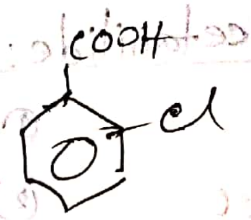
② 5% NaOH insoluble.

③ 5% HCl insoluble.

④ N/S present

So, miscellaneous neutral.

⑤ 2- chlorobenzoic acid:



142°C

① water insoluble.

② 5% NaOH soluble.

③ 5% $NaHCO_3$ soluble.

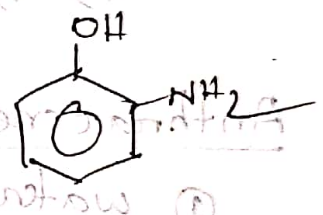
④ 5% HCl insoluble.

So strongly acidic, A_1 class.

⑥ 2-Amino phenol

174°C

① water insoluble.



② 5% NaOH soluble.

③ $NaHCO_3$ insoluble, So, A_2

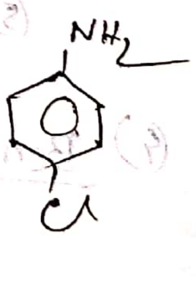
④ 5% HCl soluble, B.

So, A_2 B Amphoteric in nature.

⑦ 4- chloroaniline.

75.6°C

① water insoluble.



② 5% NaOH insoluble.

③ 5% HCl soluble.

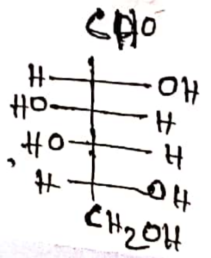
so, class B.

⑧ D (+) Glactose:

(168-170°C)

H_2O → soluble.

→ litmas paper unchange,


CHO
H
OH
HO
H
HO
H
H
OH
CH_2OH

Acetophenone:

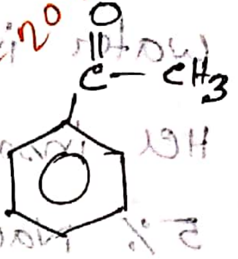
① water insoluble.

② 5% NaOH insoluble.

③ 5% HCl insoluble.

(4) conc. H_2SO_4 soluble.

(v) 85% H_3PO_4 soluble.



B.P = 202°C

M.P = 20.2°C

class-N_1.

1-heptanol:

① water insoluble.

(2) 5% NaOH insoluble.

(3) 5% HCl insoluble.

(4) conc. H_2SO_4 soluble.

(v) 85% H_3PO_4 soluble.

$CH_3-(CH_2)_6-OH$

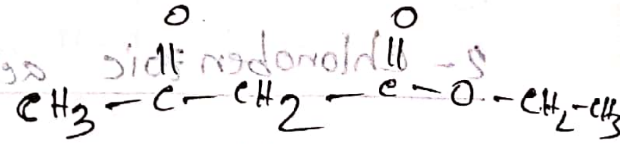
class N_1

Ethyl acetoacetate:

① water insoluble.

(2) conc. H_2SO_4 soluble.

(3) conc. H_3PO_4 soluble.



$$CH_3-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-O-CH_2-CH_3$$

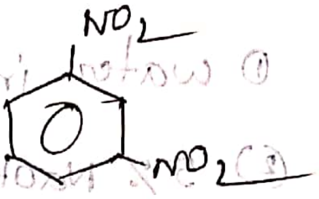
class-N_1

1,3- dinitrobenzene :

M.P = 89.6°C

① water insoluble.
② HCl insoluble.
③ 5% NaOH insoluble.
④ N present,

MN group.



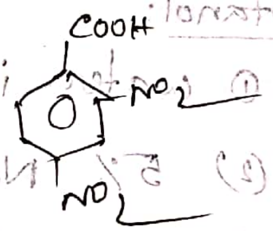
1,3- dinitrobenzene.

2,4- dinitrobenzoic acid.

187-189

① water insoluble.
② 5% NaOH soluble.
③ 5% HCl → insoluble.
④ HNO_3 – soluble.

So, A_1 class.

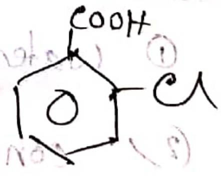


2- chlorobenzoic acid:

172°C

→ Solubility test A_1 class.

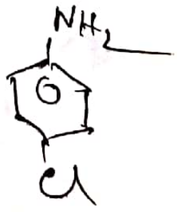
→ Halogen present. Haloacid.



4- chloro aliline : class- B

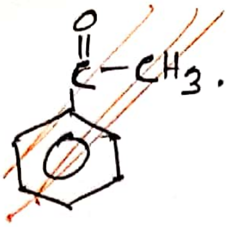
2.5°C

test for primary amine (+)

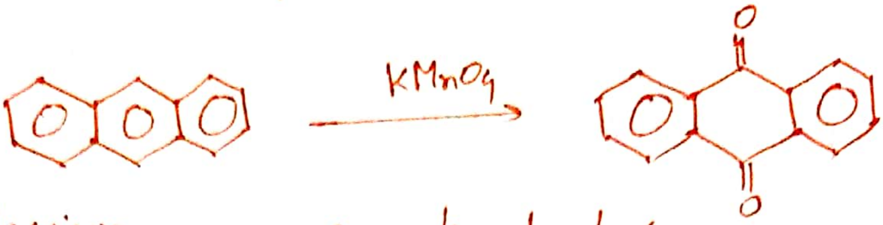


CS CamScanner

Acetophenone:



Anthracene:



→ potassium permanganate test

→ Br_2 test কোনটি দ্বারাই ধরা হয় না।

Test for carbohydrates:

(ii) (H_2SO_4 test দিলে না) white glactose H_2SO_4 add করার পর সাথেই কালো হয়)

(iv) **molisch test:**

H_2SO_4 আবার test হিসেবে দিয়ে সামনের থেকে sample দিতে হবে। violet purple junction হবে)

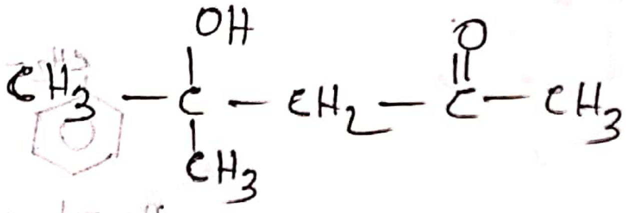
Barfoed's reagent test: সামান্য heat দিলে yellow, heat আরও দিলে red, heat কম করে দিতে হবে।

Seliwanoff's test: (Aldose test red colour আসেনি)

Rony - 26:

IR : ν_{max} 3200 – 3600 (br.), 2945, 2852, 1712, 1465, 1365 cm^{-1}

^{1}H NMR : δ 1.42 (6H, s), 2.1 (1H, br, s) 2.2 (3H, s), 2.8 (2H, s)

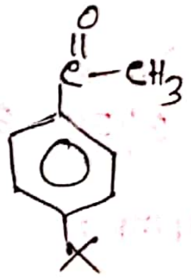


$$CH_3-\overset{\overset{\large OH}{|}}{\underset{\underset{\large CH_3}{|}}{C}}-CH_2-\overset{\overset{\large O}{\|}}{C}-CH_3$$

Rupali -150-

IR : ν_{max} 2945, 3050, 2853, 1686, 1605, 1510, 1460 cm^{-1}.

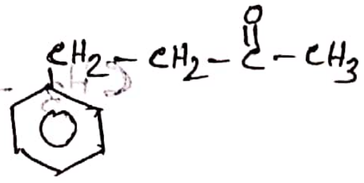
^{1}H NMR : δ 2.30 (3H, s), 7.40 (2H, d), 7.60 (2H, d)



Sadia - 161

IR: ν_{max}: 3120, 2977, 2851, 1717, 1605, 1502, 1451, 1357 cm^{-1}.

^{1}H NMR: δ 2.10 (3H, s), 2.32 (2H, t), 2.38 (2H, t), 7.36 (5H,

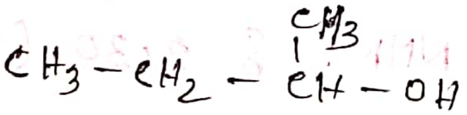


Benzylacetone

4-phenyl-2-butanone

IR: ν_{max}: 3510 - 3290 (br), 2975, 2852, 1469, 1367, 1054 cm^{-1}.

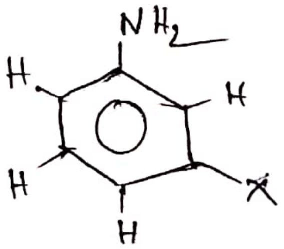
^{1}H NMR: δ 1.0 (3H, t), 1.32 (3H, d), 1.59 (2H, m)
3.41 (1H, br, s), 3.82 (1H, m).



$CH_3-CH_2-\overset{\overset{CH_3}{|}}{CH}-OH$

IR: ν_{max}:
3450, 3370, 3100, 1600, 1500, cm^{-1}.

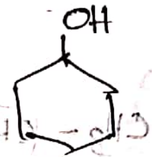
^{1}H NMR: δ 4.2 (2H, br, s), 7.3 - 7.7 (3H, m), 7.9 (1H, s)



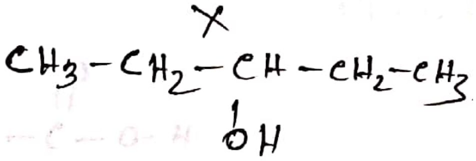
IR: ν_{max} Rakhi

3300-3500 (br.) 2930, 2865, 1460, 1050 cm^{-1};

^1H NMR: δ 1.1-1.28 (6H, m), 1.5-1.6 (4H, m), 2.85 (1H, br, s), 3.65 (1H, m).



✓ [symmetry]

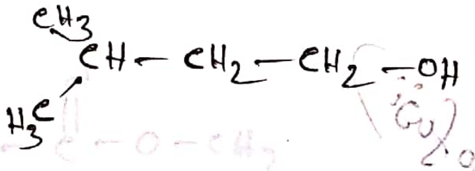


$$\underset{\textstyle OH}{CH_3-CH_2-\overset{|}{C}H-CH_2-CH_3}$$

→ (Also toma) Rakhi

IR: ν_{max} 3400-3250 (br.) 2975, 2855, 1451, 1386, and 1055 cm^{-1}

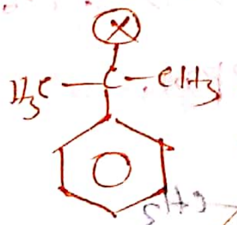
^1H NMR: δ 1.1 (6H, d), 1.5 (1H, m), 1.67 (2H, q), 3.3 (1H, br, s), 3.7 (2H, t)

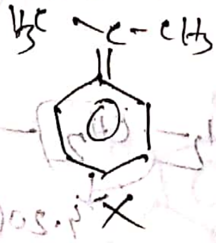


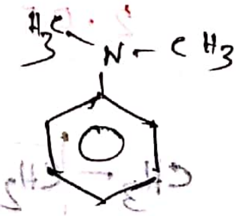
$$(CH_3)_2CH-CH_2-CH_2-OH$$

IR: ν_{max} 3071, 2960, 2840, 1603, 1506, 1466 cm^{-1}.

1H NMR: δ 2.46 (6H, s); 6.89-7.01 (5H, m)!







found

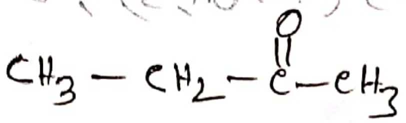
IR: ν_{max} 3201 - 3500 (br), 2942, 2850, 1466, 1368, 1056 cm^{-1}

^1H NMR: δ 0.9 (3H, t), 1.2 - 1.32 (10 H, m), 1.6 (2H, m) 2.3 (1H, br, s), 3.7 (2H, t).

$CH_3-(CH_2-CH_2-CH_2-CH_2-CH_2)-CH_2-CH_2-OH$

IR: ν_{max} 2961, 2843, 1718, 1470, 1381 cm^{-1}

^1H NMR: δ 1.1 (3H, t), 2.1 (3H, s), 2.26 (2H, q)

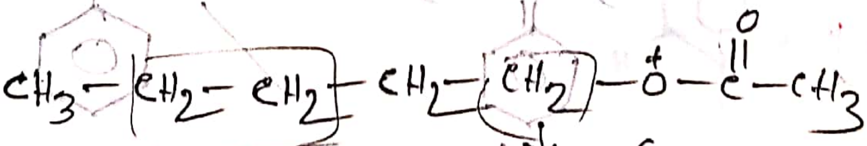


$CH_3-CH_2-\overset{O}{\overset{\|}{C}}-CH_3$

Also fuji

IR: ν_{max} 2960, 2850, 1735, 1470, 1365, 1210 cm^{-1}

^1H NMR: δ_H 0.9 (3H, t), 1.20 - 1.35 (4H, m), 1.6 (2H, m) 2.05 (3H, s), 4.20 (2H, t) ppm.

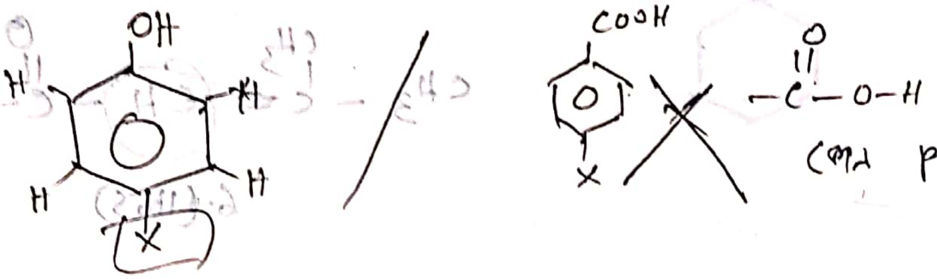


$CH_3-(CH_2-CH_2)-CH_2-\boxed{CH_2}-O-\overset{O}{\overset{\|}{C}}-CH_3$

4.20 (2H, t).

IR: ν_{max} 3600-3200 (br) 3050, 1600, 1580, 1475, 1200 cm^{-1}.

^{1}H-NMR: 6.75 (2H, d), 7.17 (2H, d), 10.25 (1H, br, s) ppm.

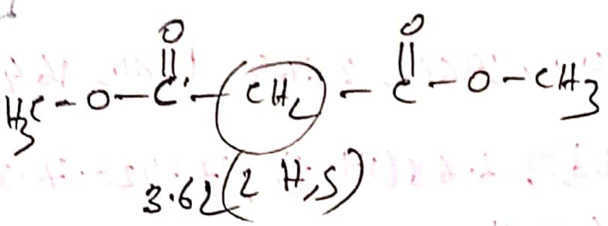

OH
H
H
H
H
X
COOH
X
O
—C—O—H

কোন peak নেই।

Sakib.

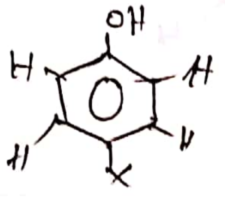
IR: ν_{max} 2966, 2860, 1740, 1472, 1366, 1201 cm^{-1}.

^{1}H-NMR: δ 3.62 (2H, s), 4.20 (6H, s)


$H_3C-O-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-O-CH_3$
3.62 (2H, s)

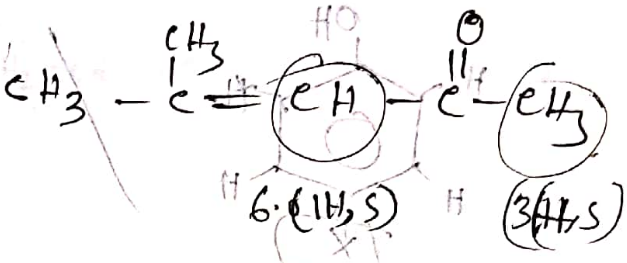
IR: ν_{max} 3600-3201 (br) 3052, 1630, 1582, 1476, 1210 cm^{-1}.

^{1}H-NMR: δ 6.70 (2H, d), 7.15 (2H, d), 10.26 (1H, br.)



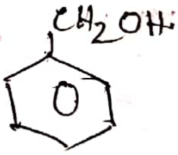
IR: ν_{max}, 3150, 2982, 2875, 1691, 1652,
1451, 1381,

¹H NMR: δ 2.1 (6H, S), 2.3 (3H, S), 6.0 (1H, S)


CH3
CH3 – C = CH – C – CH3
O
6.(1H, S)
3(H, S)

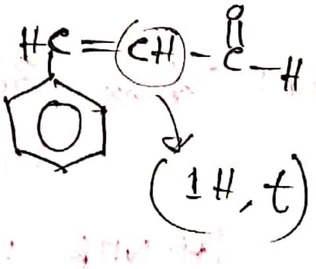
⇒ IR: 3400 – 3200 (br), 3050, 2928, 2878, 1600, 1580, 1050 cm⁻¹.

¹H-NMR: δ 2.5 (1H, br, s), 4.2 (2H, S), 7.5 – 7.8 (5H,



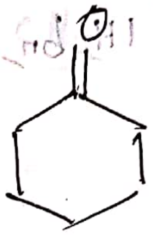
IR: 3062, 3100, 2780, 2865, 1680, 1640, 1600, 1475 cm⁻¹

NMR: 6.50 (1H, d), 6.68 (1H, t), 7.42 – 7.38 (5H, m)
9.67 (1H, d)


HC = CH – C – H
O
(1H, t)

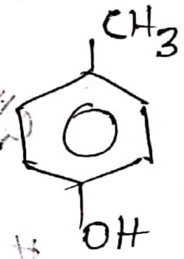
ν_{max}: 2940, 2867, 1717, 1500

NMR: 2.3 (4H, m), 1.7 (2H, m), 1.9 (4H, m).



⇒ IR: ν_{max}: 3300-3500 (br.) 3100, 2940, 2860, 1600, 1580, 1500, 1470, 1360 cm^{-1}

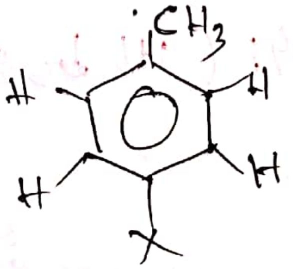
1H-NMR: δ 2.35 (3, H, S), 7.10 (H, d), 7.51 (2H, d) 10.5 (1H, br, S)



IR: ν_{max}

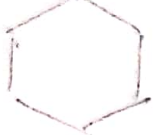
3100, 2970, 2850, 1600, 1500, 1460, 1380 cm^{-1}

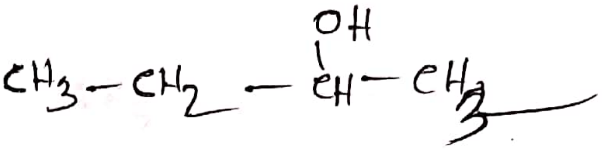
1H-NMR: δ 2.3 (3H, S), 7.3 (2H, d), 7.5 (2H, d)



ν_{max}: 3510 – 3290 (br.), 2975, 2852, 1469, 1367, 1054 cm^{-1}

^1H-NMR: δ 1.0 (3H, t), 1.32 (3H, d), 1.59 (2H, m)
3.41 (1H, br_s), 3.82 (1H, m)

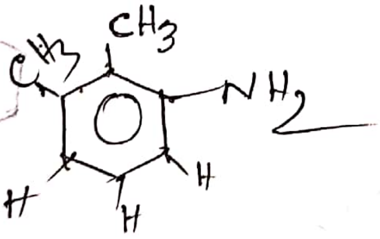




$CH_3-CH_2-\overset{OH}{\overset{|}{CH}}-CH_3$

ν_{max}: 3305, 3227, 3076, 2916, 2863, 1600, 1500, 1472 cm^{-1}

^1H-NMR: δ 2.15 (3H, S), 2.30 (3H, S), 3.48 (2H, br_s
7.09 – 7.35 (3H, m)



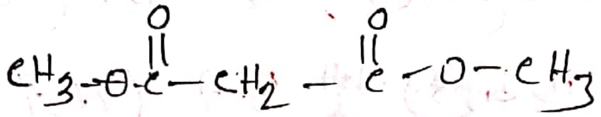
⇒ 3510 – 3290 (br.), 2975, 2852, 1669, 1367, 1054 cm^{-1}

^1H-NMR: 1.0 (3H, t), 1.32 (3H, t), 1.59 (2H, m)
3.41 (1H, br_s), 3.82 (1H, m).

(Sakib) Also

2961, 2870, 1742, 1471, 1365, 1200;

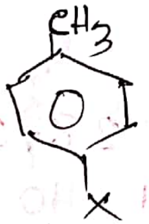
1H-NMR: 3.64 (2H, s), 4.17 (6H, s).



$CH_3-O-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-O-CH_3$

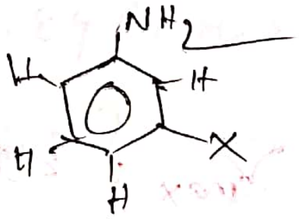
IR: ν_{max} 3050, 2928, 2850, 1600, 1515, 1442 cm⁻¹.

1H-NMR: δ 2.28 (3H, s), 6.9 (2H, d), 7.3 (2H, d)



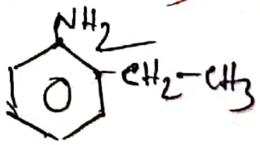
IR: ν_{max} 3450, 3370, 3100, 1600, 1500, cm⁻¹.

1HNMR: δ 4.2 (2H, br, s), 7.3–7.7 (3H, m), 7.9 (1H, s).



IR: ν_{max} 3405, 3390, 3040, 2990, 2850, 1600, 1580, 1500, 1470, 1380, cm⁻¹.

1H-NMR: δ 1.3 (3H, t), 2.3 (2H, q), 4.5 (2H, br, s)
7.10–7.25 (4H, m), ppm.



IR: ν_{max}: 3500 – 3400 (br.), 2965, 2850, 1470, 1360, 1050 cm^{-1}

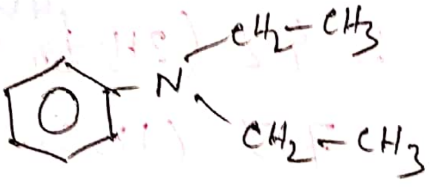
^1H-NMR: δ 0.9 (3H, t), 1.2 – 1.35 (8H, m), 1.6 (2H, m), 2.2 (1H, br, s), 3.7 (2H, t) ppm.

- $$CH_3-\underbrace{(CH_2-CH_2-CH_2-CH_2)}_{(8H,m)}-CH_2-CH_2-OH$$

IR: ν_{max}:

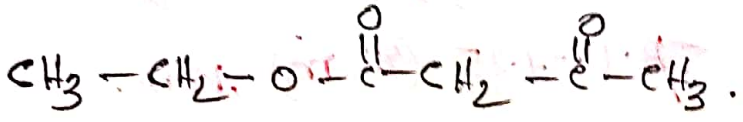
3100, 2930, 2875, 1601, 1580, 1500, 1470, 1365 cm^{-1}

^1H-NMR: δ 1.6 (6H, t), 2.5 (4H, q), 7.2 – 7.3 (5H, m).



IR: ν_{max}: 2965, 2855, 1745, 1723, 1465, 1360, 1230 cm^{-1}

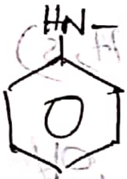
^1HNMR: δ 1.45 (3H, t), 2.0 (3H, s), 4.1 (2H, q), 4.5 (2H, s).



$$CH_3-CH_2-O-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-CH_3.$$

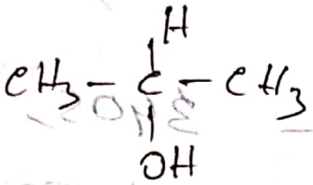
IR: 3250, 3160, 2920, 2855, 1610, 1582, 1500, 1481, 1366 cm⁻¹

¹H-NMR: δ 2.3 (3H, s), 4.2 (1H, br s), 7.3-7.4 (5H m



νmax: 3200-3600 (br), 2965, 2860, 1220 cm⁻¹.

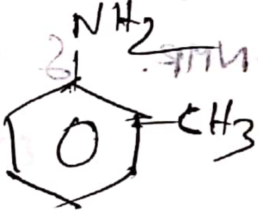
¹H-NMR: 5.1 (H, s, br), 3.98 (1H, m), 1.11 (6H, d).



$CH_3-\overset{H}{\underset{OH}{C}}-CH_3$

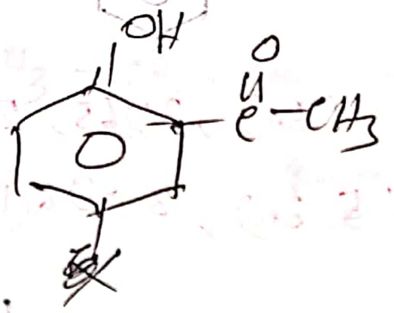
IR: 3432, 3346, 3050, 2924, 2853, 1600, 1510, 1472, 1302, cm⁻¹.

¹H-NMR: δ 2.19 (3H, s), 3.84 (2H, s, br.), 7.10-7.35 (4H, m).



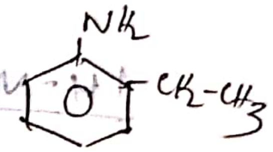
IR: ν_{max}: 3300 – 3450 (br.), 3100, 2950, 2865, 1685, 1600, 1580, 1500, 1480, 1365 cm^{-1}.

^{1}H NMR: δ 2.30 (3H, s), 7.2 – 7.4 (4H, m), 10.55 (1H, br, s).



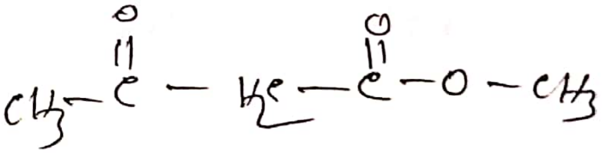
IR: **ν_{max}:** 3405, 3390, 3040, 2940, 2850, 1600, 1580, 1500, 1470, 1380 cm^{-1}.

^{1}H-NMR: δ 1.3 (3H, t), 2.3 (2H, q), 4.5 (2H, br, s), 7.10 – 7.25 (4H, m).



IR: **ν_{max}:** 2969, 2849, 1748, 1718, 1439, 1362, 1130 cm^{-1}.

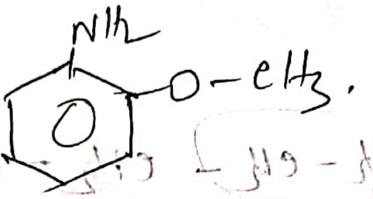
^{1}H-NMR: δ 2.20 (3H, s), 4.60 (3H, s), 4.40 (2H, s).



$CH_3-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-O-CH_3$

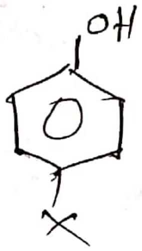
IR: ν_{max}: 3400, 3280, 3080, 2940, 2850, 1610, 1500, 1480, 1360, 1200 cm^{-1}

^1H-NMR: δ 3.5 (2H, br, s), 3.85 (3H, s), 6.7 − 7.3 (4H, m) ppm.



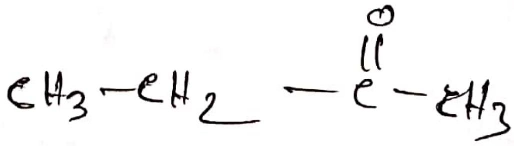
IR: ν_{max} 3600−3200 (br, s) 3050, 1600, 1580, 1475, 1200 cm^{-1}.

^1H-NMR: δ 6.75 (2H, d), 7.17 (2H, d), 10.25 (1H



IR: ν_{max} 2960, 2842, 1715, 1470, 1380 cm^{-1}.

^1H-NMR: δ 1.1 (3H, t), 2.1 (3H, s), 2.25 (2H, q) ppm



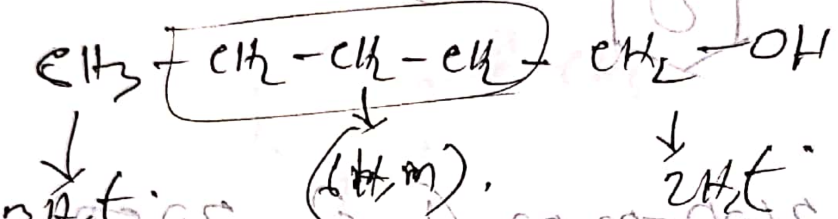
$CH_3-CH_2-\overset{O}{\overset{\|}{C}}-CH_3$

ν_{max}: 3400 – 3200 (br.), 2970, 2850,

1450, 1385, 1060 cm^{-1}.

1H-NMR: 1.0 (3H, t), 1.6 (6H, m), 3.2 (1H, br, s)

3.6 (2H, t).



$CH_3-CH_2-CH_2-CH_2-CH_2-OH$